

Singapore HDB Resale Trend Analysis Using Python

Project Objective:

- To track town by town price growth trajectories.
- To analyse the property valuation over the years.
- Analyze the resale price index to quantify.
- Evaluate the flat type influence the resale.
- To analyse the remaining lease period impact on property value.
- To identify how the flat model get impacted for resale.

Outcome:

This analysis concludes the better understanding of singapore HDB resale data through exploratory data analysis and visualization techniques. This project helps to discover the trends and relationships over the HDB resale data.

Dataset Information:

Source:

[https://data.gov.sg/datasets?topics=housing&resultId=d_8b84c4ee58e3cfc0ece0d773c8ca6abc]

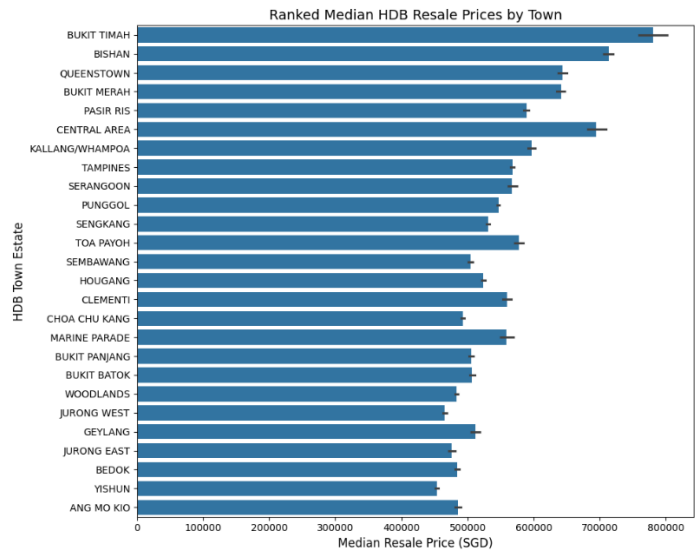
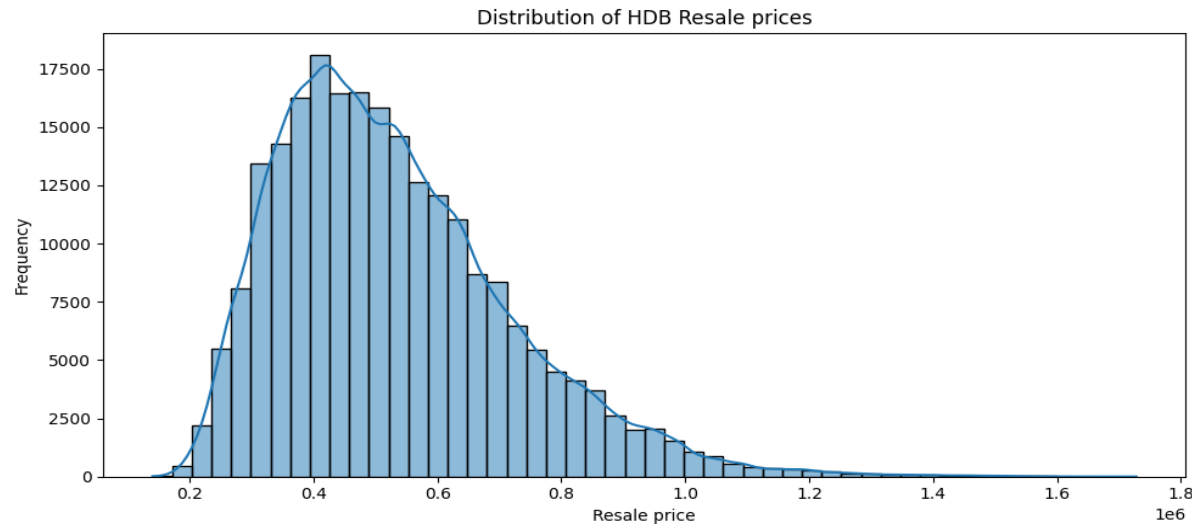
Dataset Column/Feature Description:

Column Name	Data Type	Description
month	Date	The year and month when the resale transaction was registered (formatted as YYYY-MM).
town	Text	The specific HDB estate or residential town where the property is located.
flat_type	Text	The room type configuration of the HDB unit (e.g., number of rooms or executive status).
block	Text	The specific block number of the HDB building.
street_name	Text	The official street name where the HDB building is located.
storey_range	Text	The grouped floor level range where the specific unit is located.
floor_area_sqm	Numeric	The total internal floor area of the flat, measured in square meters.

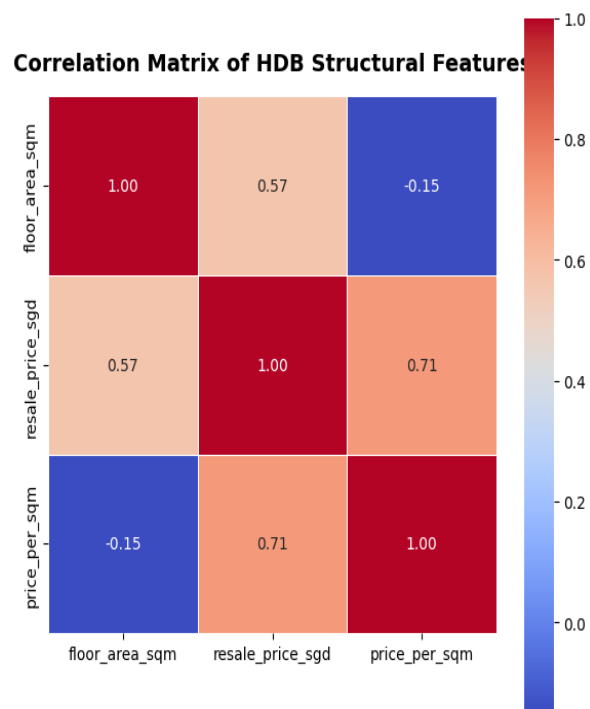
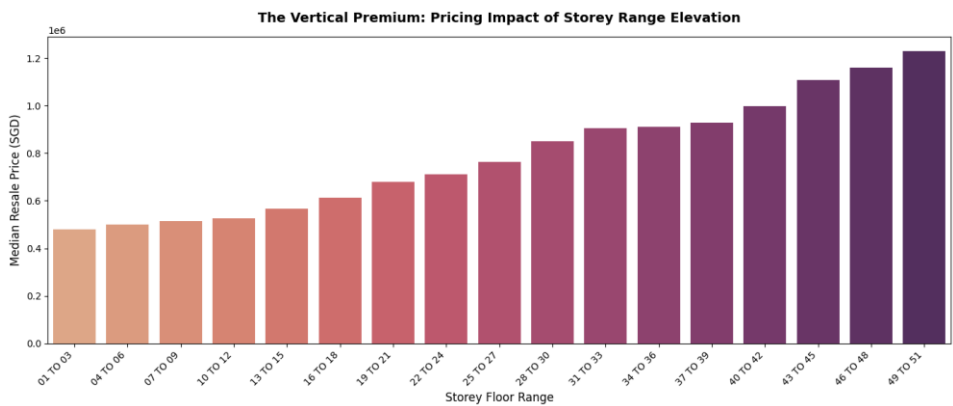
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Column Name	Data Type	Description
flat_model	Text	The design model or architectural classification of the HDB flat.
lease_com_mence_date	Year	The year the 99-year lease officially began for the building.
remaining_l ease	Text	The remaining duration of the 99-year lease at the time of the transaction (expressed in years and months).
resale_price	Num eric	The final agreed-upon sale price of the flat in Singapore Dollars (SGD).

EDA AND VISUALIZATION:

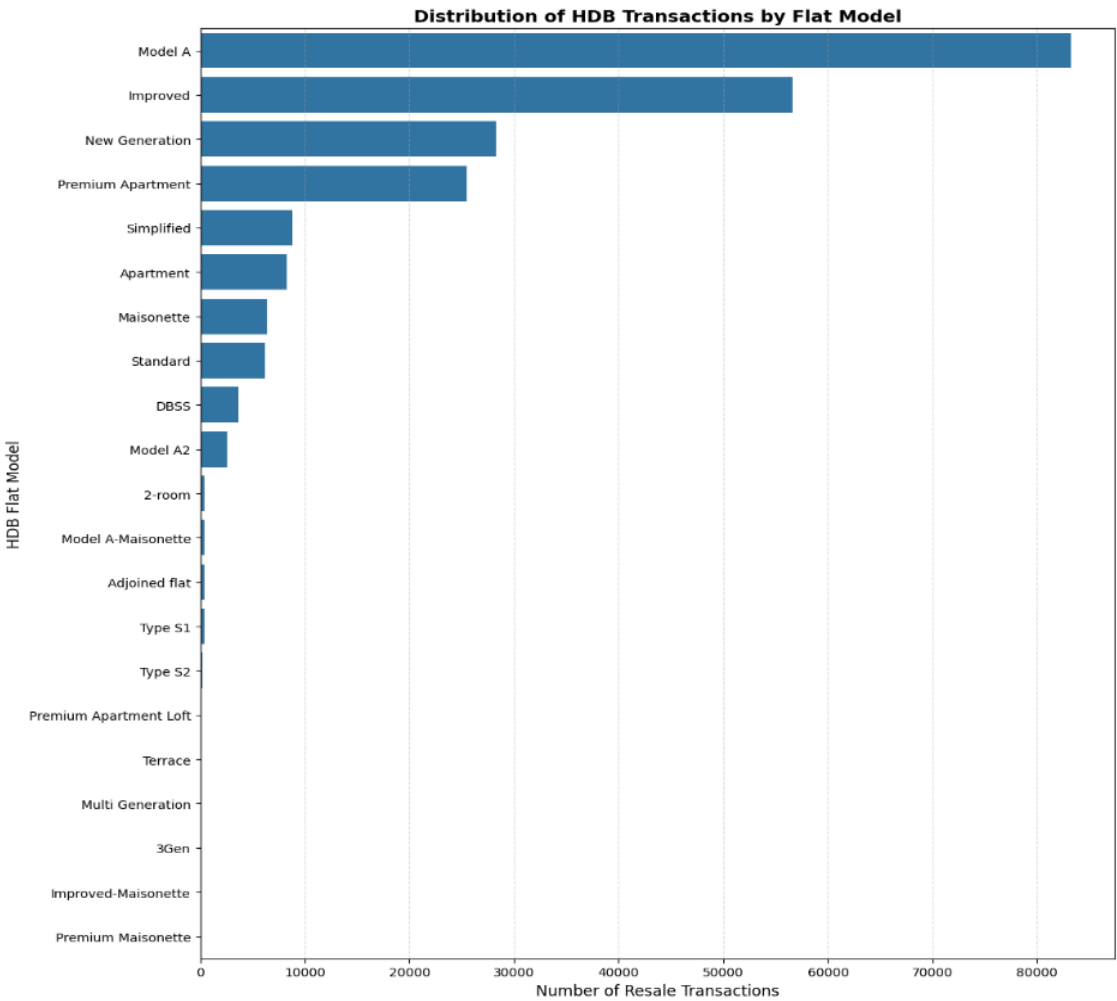
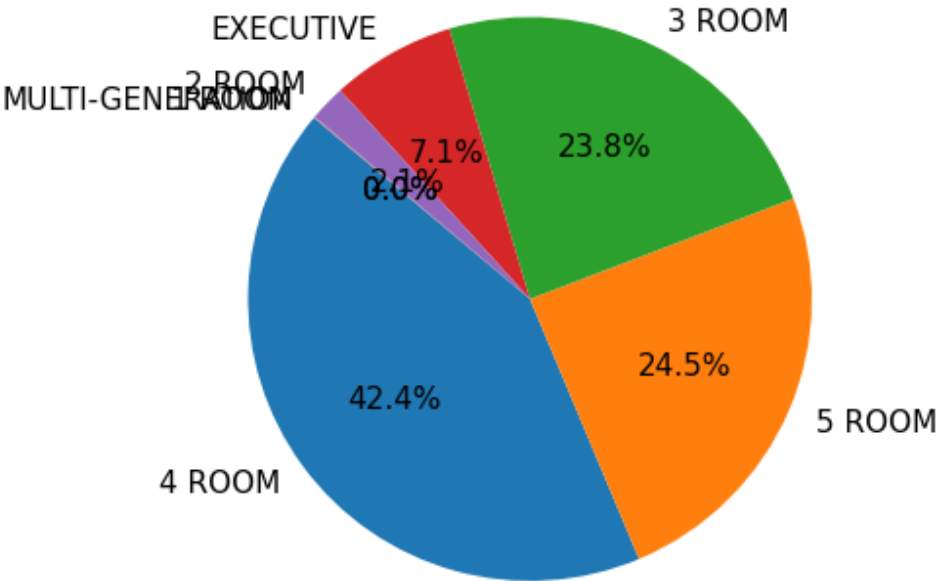


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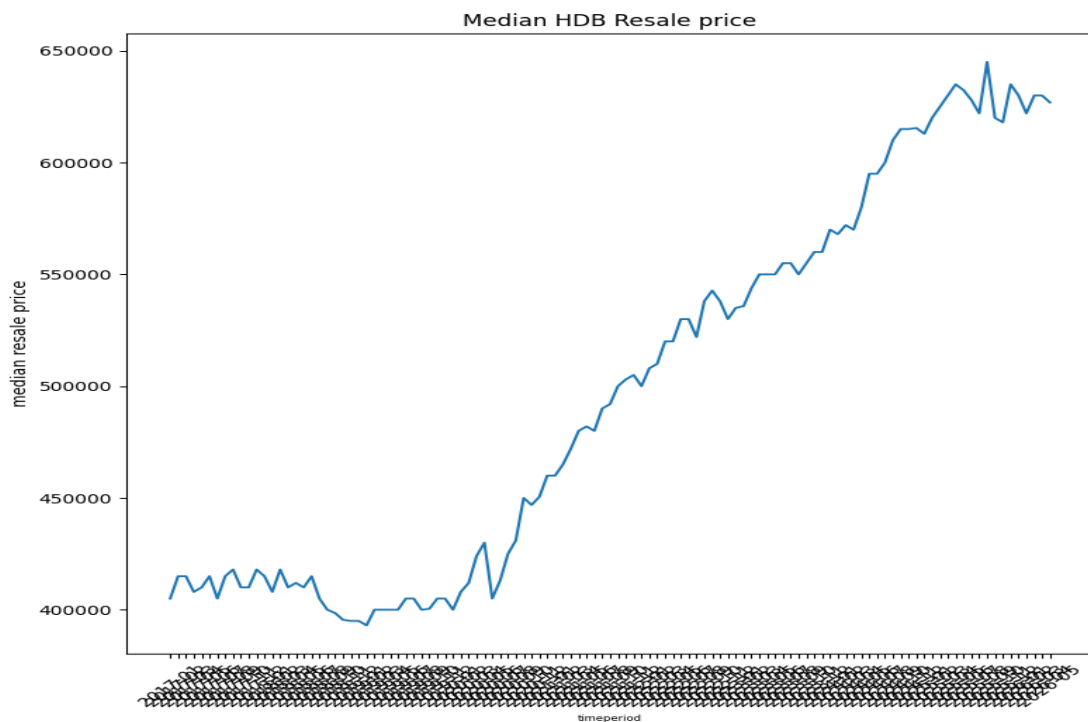
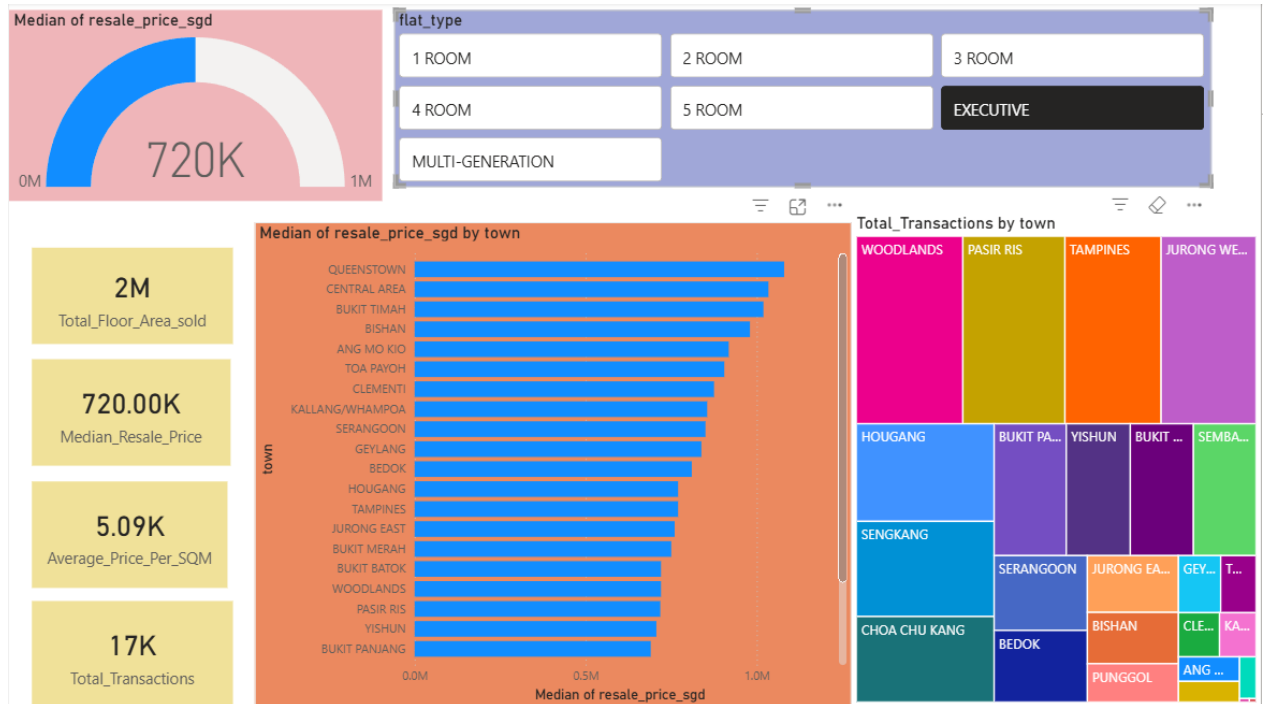


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Market Share Distribution of HDB Flat Types



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Insights:

- HDB resale prices clearly breakouts the profound impact of the pandemic on real estate.
- The vast majority of HDB flat transactions occurs affordable, entry to mid tier price points. However the market has significant premium tier pulling the average upward.
- premium buyers place on central proximity, prestige, and top-tier schooling districts. Even though HDB flats are public housing, a prime location like Bukit

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Timah or Bishan drives median transaction costs significantly higher than suburban equivalents.

- It demonstrates that physical size is not the main driver of high-end property valuation in Singapore, location is. A 90 sqm 4-room flat in a non-mature, far-flung estate sits at the lower end of that vertical line, while an identical 90 sqm 4-room flat located in a prime central area (like Dawson or Queenstown) shoots up to the top of the line. Location premiums completely override raw square footage.
- The floor height acts as a major premium multiplier in Singapore's public housing sector. Buyers are systematically willing to pay a progressive surplus for higher units. The premium is driven by three main value factors: better panoramic views, reduced road and street-level noise, and superior ventilation/natural light.
- 4-room units are the definitive "standard baseline" of Singaporean public housing.
- Flats built in the late 1970s, 1980s, and 1990s were heavily dominated by "Model A" and "Improved" architectural blueprints. Because these units represent the primary housing supply ever built across major heartland mega-towns, they form the bedrock liquidity of today's resale market

Conclusion:

- This Singapore HDB Resale Trend Analysis project successfully bridges raw historical property records with interactive business intelligence.
- The Singapore public housing market is no longer a uniform utility—it behaves like a mature, competitive asset class. While suburban towns like Woodlands and Sengkang ensure foundational market volume and baseline affordability, a hyper-competitive layer exists where premium location models (like DBSS or high-storey central units) behave like luxury assets.